Protected cycle tracks for Avinguda del Paral·lel

Group report – PACE 1, Project B: bike infrastructure integration

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Summary

Avinguda del Paral·lel already has a bike lane, yet on our visit most cyclists were not using it. The lane runs down the middle of the avenue, it is 2.2 m wide for both directions, and it leaves riders with moving traffic on either side. People prefer to ride along the kerb, next to parked cars, where the shops, hotels and the metro are. We took that behaviour as our starting point.

On a 365 m stretch around the Ronda de Sant Pau junction, we propose to remove the central lane and build a 2.0 m one-way cycle track along each kerb. Each track is protected by a 0.4 m buffer and a 2.1 m parking and loading strip, so parked cars, not a painted line, separate cyclists from traffic. The bus lanes, the general traffic lanes and the parking supply stay as they are. Our six indicators show that space for cycling almost doubles and the main conflicts disappear; the price is a narrower sidewalk zone and some new conflicts at bus stops and side streets, which the design addresses directly.

The project at a glance

Question	Our answer
Where?	365 m of Av. del Paral·lel, C. del Comte Borrell to C. Nou de la Rambla, including the Ronda de Sant Pau junction (metro Paral·lel).
Why here?	The existing central lane is narrow (1.1 m per direction), poorly connected and avoided by many cyclists, on a busy 3-lane-per-direction avenue.
What?	A 2.0 m one-way cycle track along each kerb, protected by a 0.4 m buffer and a 2.1 m parking/loading strip, with floating bus stops and a protected junction.
Why this design?	It follows where cyclists already ride, gives physical separation on a 50 km/h road (CROW) and keeps bus lanes, traffic lanes and parking.
Does it work?	Six indicators: cycling space almost doubles, cyclists stop riding between two traffic streams, and the walking path widens; the sidewalk zone narrows.

1. Introduction and objectives

The brief asked each group to pick a real street in Barcelona, to diagnose it and to develop – and justify – a cycling intervention on a corridor of 300–800 m that includes at least one relevant junction. As the course put it, the task was not to find the correct solution but to defend a reasoned one. We followed the common method of the course: select, diagnose, design, evaluate and justify, using the CROW manual and the Barcelona and Catalan design guides as our main evidence.

We set ourselves three objectives, which we use again in the evaluation:

- **Safety:** cycling that is physically separated from moving traffic.
- **Connectivity:** direct access from the bike route to destinations, side streets and the wider network.
- **Comfort:** at least 2 m of cycling space per direction, without taking away bus lanes, traffic lanes or parking.

The people who gain most are everyday cyclists and those who would like to cycle but feel unsafe today – the "interested but concerned" group that Geller (2006) estimates at around half of the population – together with Bicing users, metro and bus passengers, and the businesses along the avenue.

2. Location

Our segment runs for 365 m along Av. del Paral·lel, from Carrer del Comte Borrell to Carrer Nou de la Rambla. Near its middle is the junction with Ronda de Sant Pau and C. d'En Fontrodona, right above the Paral·lel metro station (lines L2 and L3) and beside the small public square of Plaça de la Bella Dorita. The length was measured in Google Maps and sits comfortably inside the 300–800 m range of the brief.

We chose the place for a simple reason: it is a street that already has cycling infrastructure, but infrastructure that does not work well. The central lane is hard to join from the side streets and unpleasant to ride, with cars on both sides. At the same time, the street offers real room to work with – a wide median, very wide sidewalks, a metro station, a public square and bike connections in every direction.

The brief describes Project B as introducing cycling infrastructure where none exists. Strictly, Paral·lel has a lane; in practice, the segment lacks infrastructure that cyclists can use safely and comfortably, which is why we treat it as a case of integrating proper cycling infrastructure rather than repainting what is there.

3. Diagnosis of the existing street

Paral·lel is a broad, historic avenue that links Plaça d'Espanya with the waterfront at the Columbus Monument. It also works as a border: El Poble-sec lies to the south, Sant Antoni and El Raval to the north. In our segment the avenue is a busy hub. The metro entrances, bus stops and taxi ranks sit next to theatres and clubs such as Sala Apolo and Teatre Victòria, hotels such as the Barcelona Universal, and a row of restaurants and take-aways, so pedestrian activity is high during the day and late into the night.

Building to building, the street is about 40 m wide. Table 1 gives the existing allocation of that space, as measured on satellite imagery and Street View and redrawn as a scaled section.

Element (north to south)	Width (m)	Element (continued)	Width (m)
Sidewalk, with shelters, trees, terraces	7.2	Divider	0.3
Parking strip	2.0	2 traffic lanes	5.3
Bus lane	3.0	Bus lane	3.0
2 traffic lanes	5.3	Parking strip	2.0
Planted median	1.0	Sidewalk, with shelters, trees, terraces	8.7
Two-way bike lane	2.2	**Total**	**40.0**

Table 1. Existing cross-section of Av. del Paral·lel in the study segment (approximate).

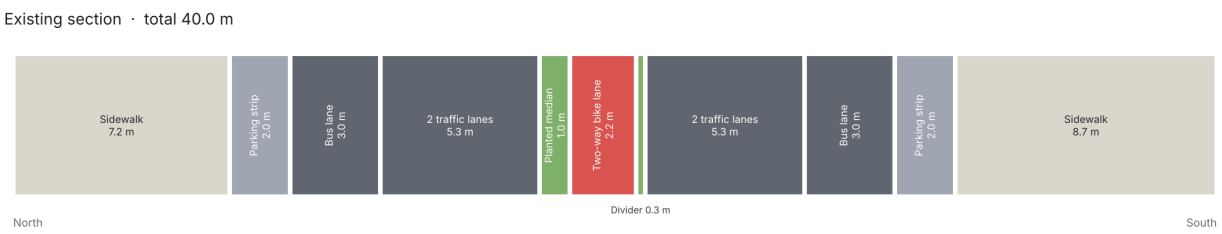


Figure 1. Existing cross-section, building to building (approximate widths).

Walking the segment, several problems stood out:

- **The lane is not used as intended.** Many cyclists ride near the road shoulder instead of in the central lane.

- **It is too narrow.** 2.2 m for two directions leaves 1.1 m per direction, well below the 2.50 m CROW recommends even for a low-volume two-way path.
- **It feels exposed.** Riders have moving traffic on both sides, on a road with three lanes per direction where cars can drive quite fast.
- **It is poorly connected.** A lane in the middle of the road does not lead directly to side streets, shops or the metro.
- **The kerb is cluttered.** Cars, motorbikes and rubbish bins line the shoulder.

Some things had to stay. Bus lanes and stops, the metro entrances, the street trees, access for loading and taxis, and the avenue's role in the traffic network were treated as fixed. Our room for change lay in the median, in the central lane itself and in the organisation of the sidewalks.

4. Proposed design

We replace the central two-way lane with two one-way cycle tracks, one along each kerb and in the same direction as the traffic beside it. The order of elements on each side, from the building line, becomes: sidewalk zone, cycle track, buffer, parking and loading strip, bus lane and two traffic lanes. In other words, the cycle track moves to the inside of the parking strip, and parked cars become the barrier between cyclists and moving traffic.

Element (per side)	Before (m)	After (m)
Sidewalk zone	7.2 / 8.7	6.7 (3.3 clear path + trees, terraces or benches)
Cycle track	–	2.0, one-way
Buffer	–	0.4
Parking / loading strip	2.0	2.1
Bus lane	3.0	3.0
2 traffic lanes	5.3	5.3 (2 × 2.65)
Central bike lane + divider (shared)	2.2 + 0.3	removed
Median (shared)	1.0	1.0, planted

Table 2. Space allocation before and after the intervention (typical section, total 40 m).

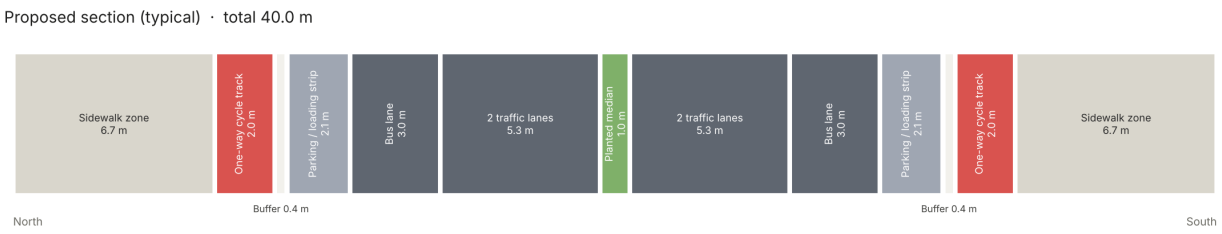


Figure 2. Proposed typical cross-section.

We kept the 0.4 m buffer on purpose. Next to a parking strip, it is what keeps a cyclist out of the way of a car door opening, and it matches the minimum safety band in the Barcelona manual (fiche T-2a). We discussed using that space to widen the track, and decided that the buffer was worth more.

Bus stops. At each stop the buffer and parking strip are replaced by a 2.7 m floating platform with the shelter. The track runs behind the platform, so buses never cross the cyclists' path, and passengers reach the platform over a zebra crossing (Barcelona manual, fiche Y-9a).

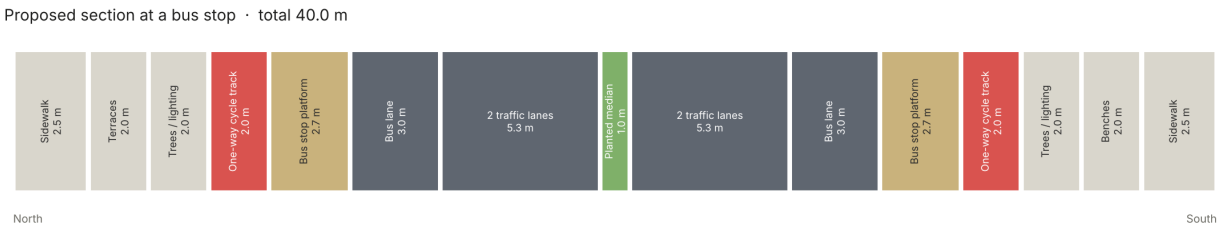


Figure 3. Proposed cross-section at a bus stop, with floating platforms.

The Ronda de Sant Pau junction. The junction is where most of the risk sits: more than 70% of urban pedestrian and cyclist collisions happen at intersections. We designed it as a protected junction. The tracks continue across each arm as red cycle crossings next to the zebras; stop lines for cars are set back before every crossing; cyclists get their own signal heads with advance detection; and right turns into the side streets are free for cyclists. At the quiet side street of C. d'En Fontrodona the track stays level across the entrance and cars give way.

The team drew a full plan of the junction; its main elements are listed below.

Element of the junction	What it does
Red one-way cycle tracks (2.0 m) on both sides	Continue through the junction in the same direction as traffic, with signs where the track starts and edge markings where it bends.
Raised red cycle crossings beside each zebra	Carry cyclists straight across each arm; cars are held on red while cyclists cross, so the crossing is conflict-free.
Two-way cycle crossing, 3.0 m, with a 2.0 m waiting space	Lets cyclists change side of the avenue in one signal phase.
Stop lines set back before every crossing	Cars stop before the cycle crossing, which keeps cyclists visible to drivers.
Cycle signal heads with advance detection	Cyclists get their own green, triggered on request; the signal cycle is at most 90 s, with a bike green twice per cycle.
Free right turns for cyclists	Cyclists turning into Ronda de Sant Pau or C. d'En Fontrodona do not have to stop at the signal.
Raised exit at C. d'En Fontrodona	The track stays level across the side street; cars give way, with shark's teeth and a give-way sign.
Floating bus stops on both sides	Platforms in the protection zone; the track passes behind the shelter and passengers board over a zebra.
Metro entrance and Bicing docks kept	Existing access points stay in place, clear of the track.
Planted median (1.0 m)	Replaces the old central lane as a planted strip; car signal heads stand at every stop line.

Table 3. Main elements of the proposed protected junction.

Connections and other uses. Cyclists join and leave the corridor at every signalised junction, which links the tracks to bike routes in each direction. Nothing is taken from buses or general traffic, and the parking and loading strip is kept. The one real loss is width in the sidewalk zone, which narrows from 7.2 and 8.7 m to 6.7 m on each side. We reorganised it so that the clear walking path actually grows, from 2.7 m to 3.3 m, with trees, benches and terraces in a separate band.

5. Evaluation

We compared the existing street with our design using six indicators, chosen so that each of the course's categories – mobility, safety and accessibility, urban quality, and traffic and feasibility – is covered at least once.

Indicator	Existing	Proposed	What it means
Cycling width per direction (mobility)	1.1 m (2.2 m two-way lane)	2.0 m one-way, each side	About 80% more space per direction; meets CROW (2.0 m for up to 150 bikes/h) and Barcelona fiche T-2a.
Separation from moving traffic (safety)	0.3 m divider and 1.0 m median; traffic on both sides	0.4 m buffer and 2.1 m parking strip; traffic on one side	Cyclists no longer ride between two traffic streams on a 50 km/h road.
Traffic lanes crossed to reach the kerb (access)	3, plus the parking strip	0 on the same side	Shops, hotels, theatres and the metro can be reached directly.
Clear walking path (urban quality)	2.7 m	3.3 m, plus trees, benches, terraces	A wider path for walking, although the overall sidewalk zone is narrower.
Share of the 40 m (space allocation)	Walk 40% · bike 5.5% · traffic 41.5% · parking 10% · median 3%	Walk 33.5% · bike 10% (+2% buffer) · traffic 41.5% · parking 10.5% · median 2.5%	Space for cycling almost doubles; traffic and parking are unchanged.
Bus stops and kerb uses (traffic & feasibility)	Stops at the kerb; 2.0 m parking strip	Floating stops (2.7 m); 2.1 m strip kept	No bus–bike conflict at stops; passengers cross the track on a zebra.

Table 4. Before/after comparison. Widths are approximate and come from our Streetmix sections (satellite imagery, Street View and a 40 m total); lanes crossed were counted on the sections.

The design does well on the three objectives we set. On safety, cyclists move from a narrow strip between two traffic streams to a track with traffic on one side only, behind a row of parked cars. On connectivity, the route now touches every shop front, side street and metro entrance. On comfort, each direction gets 2.0 m, which lets two people ride side by side or overtake.

It also has costs, and we want to name them plainly. The sidewalk zone loses between 0.5 and 2.0 m. The 0.4 m door buffer is the minimum the Barcelona manual allows, not a generous margin. Bus passengers now cross the cycle track to reach the platform. And because the tracks run along the kerb, cyclists meet turning cars at every side street, which is why the junction design matters as much as the cross-section.

6. Alternatives and justification

We weighed our proposal (Option A) against the most obvious alternative: keeping the central two-way lane, widening it to 3.0 m by using the median, and adding protected signals at the junctions (Option B). Table 5 sets out how we judged them on the six criteria of the course.

Criterion	Option A – kerbside one-way tracks	Option B – wider central lane
Safety	++	+
Connectivity	++	0
Comfort	+	0
Urban quality	+	0
Traffic impact	0	0
Feasibility	–	+

Table 5. Multi-criteria comparison (++ very positive, + positive, 0 neutral, – negative).

Option B is cheaper and simpler to build, and it would make the existing lane more comfortable. But it leaves the two problems we saw on site untouched: riders would still have traffic on both sides, and they would still have to cross three lanes to reach anything on the street. Option A costs more – kerbs, signals and bus stops have to be rebuilt – but it answers the problems directly. Four arguments decided it for us:

1. **It follows what people already do.** Cyclists on Paral·lel ride at the kerb today. A design that puts the track there works with that behaviour instead of against it.

2. **It solves the problems from our diagnosis.** Kerbside tracks connect to every side street and destination, and they remove the feeling of riding between two flows of cars.
3. **The road calls for separation.** With three lanes per direction and speeds of up to 50 km/h, CROW points to separate cycle tracks rather than lanes or mixed traffic; a speed difference above about 30 km/h needs physical separation.
4. **It matches the course's worked example.** For a main urban distributor with shops on both sides, the solution chosen in class was one-way cycle tracks on each side (CROW sheet V15).

The widths come from the same sources. CROW gives 2.0 m for a one-way track carrying up to 150 cyclists per hour, and the Barcelona manual's fiche T-2a combines a 2.0 m one-way lane with a 0.4 m safety band – the dimensions we used.

7. Limitations and next steps

Our proposal rests on assumptions that would need checking before anyone built it:

- All widths are approximate. They come from satellite imagery, Street View and Streetmix, not from a measured survey.
- We have no cyclist counts. If peak flows are above 150 bikes per hour, CROW recommends 2.5–3.0 m tracks, and the layout would need adjusting.
- We have no data on vehicle speeds, traffic volumes or parking and loading demand.
- Our Lane Patrol (CycleRAP) assessment covered Bicivia 7 between Av. Països Catalans and Av. Diagonal, not this segment, so we could not give a CycleRAP risk score before and after.
- The junction drawing is schematic. Signal timings and turning movements would need a proper traffic study.

As the course reminded us, a weak first result does not always mean an intervention has failed: the network may still be disconnected, or the main barrier may remain at a junction. That is why we would not stop at construction. Our next steps would be to count cyclists at peak hour, to record the segment in Lane Patrol and run a CycleRAP forecast for the new layout, and to test the design first as a temporary, tactical-urbanism trial. Data collected before and after the trial, and compared with a similar control street, would show whether the design works and where it needs to change.

8. Conclusion

Paral·lel does not lack a bike lane; it has one in the wrong place. Moving the cycling space from the centre of the road to the kerb, and protecting it with a buffer and a row of parked cars, gives every direction 2 m of separated space, connects the route to the street's shops and metro, and does so without removing a single bus lane, traffic lane or parking strip. The trade-offs are real but manageable, and the next step is clear: measure, test it on the street, and adjust.

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